

SYSTEM ARCHITECTURE :

Introduction:

The proposed System Architecture is a multi-tier Cyber-Physical Power System (CPPS) designed to ensure that decentralized energy trading remains within safe operational limits while staying resilient to cyber-attacks. The architecture is divided into four distinct functional layers coordinated by a centralized control system

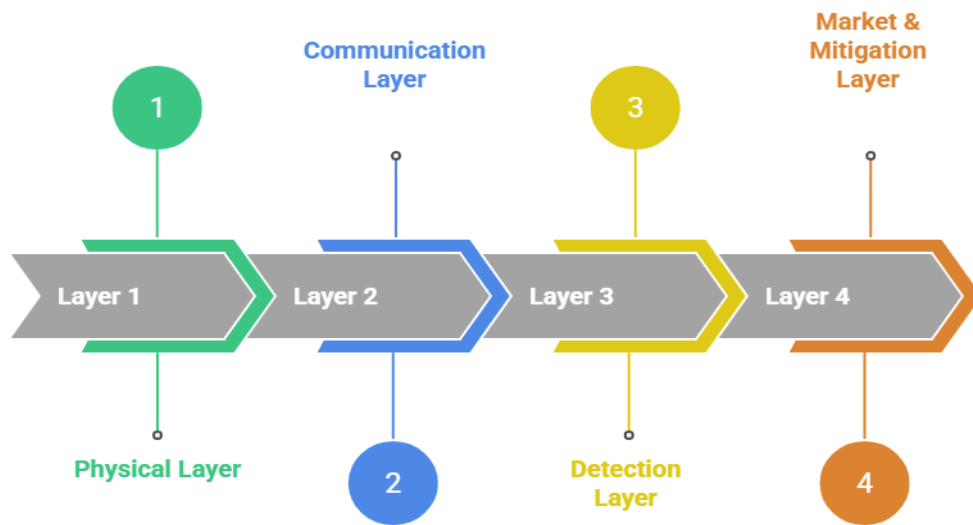


Figure: System architecture

The project follows a systematic cyber-physical approach, divided into four core phases: Physical Layer Setup, Communication layer, and detection, market & mitigation layer. The detailed system Architecture is shown below fig 2.1.1 where each layer sub classification is done. The Physical Layer based on the modified IEEE 13-bus distribution feeder simulated in OpenDSS, incorporating three DER prosumer nodes and consumer load nodes, returning per-bus voltage magnitudes, active power injections, and reactive power injections at each measurement cycle. The Communication Layer is implemented through Raspberry Pi-based RTUs and smart meters that collect grid measurements, add calibrated sensor noise, encrypt the measurement vector using blockchain-retrieved key parameters, and transmit two independent streams over TCP. The Control Centre Layer consists of Python Flask servers forming the MCO master controller handles SCADA functions including node authentication, WLS bad-data detection, FCI command validation, and OpenDSS voltage verification,

alongside the RL-SDN DoS mitigation agent. The Market and Mitigation Layer is governed by smart contract which manages P2P double auction settlement, encryption key rotation, and role-based device authentication with on-chain node revocation ensuring that any detected compromised node is immediately isolated from both the measurement stream and future energy trading without requiring manual operator intervention.

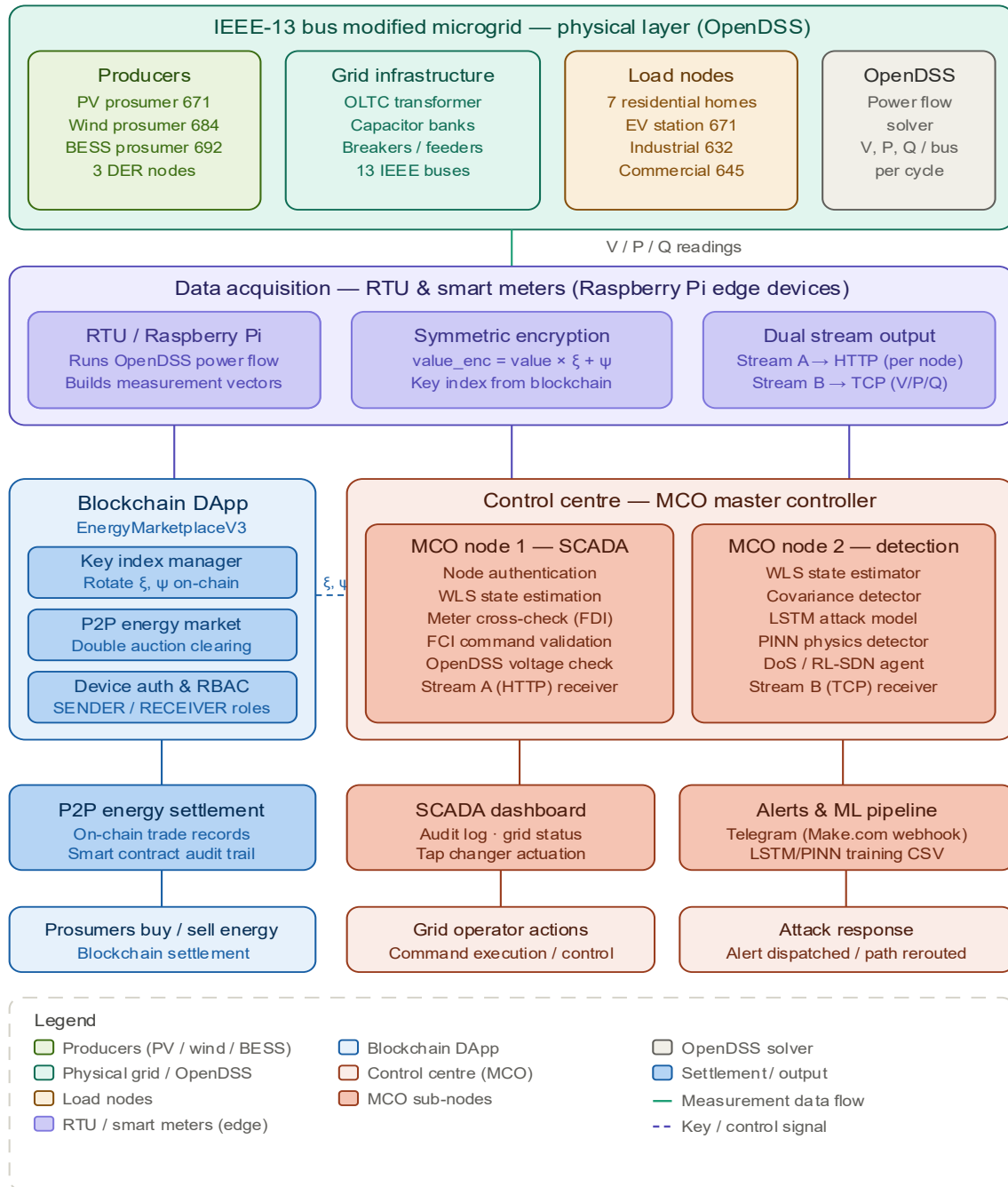


Figure: Detailed system architecture

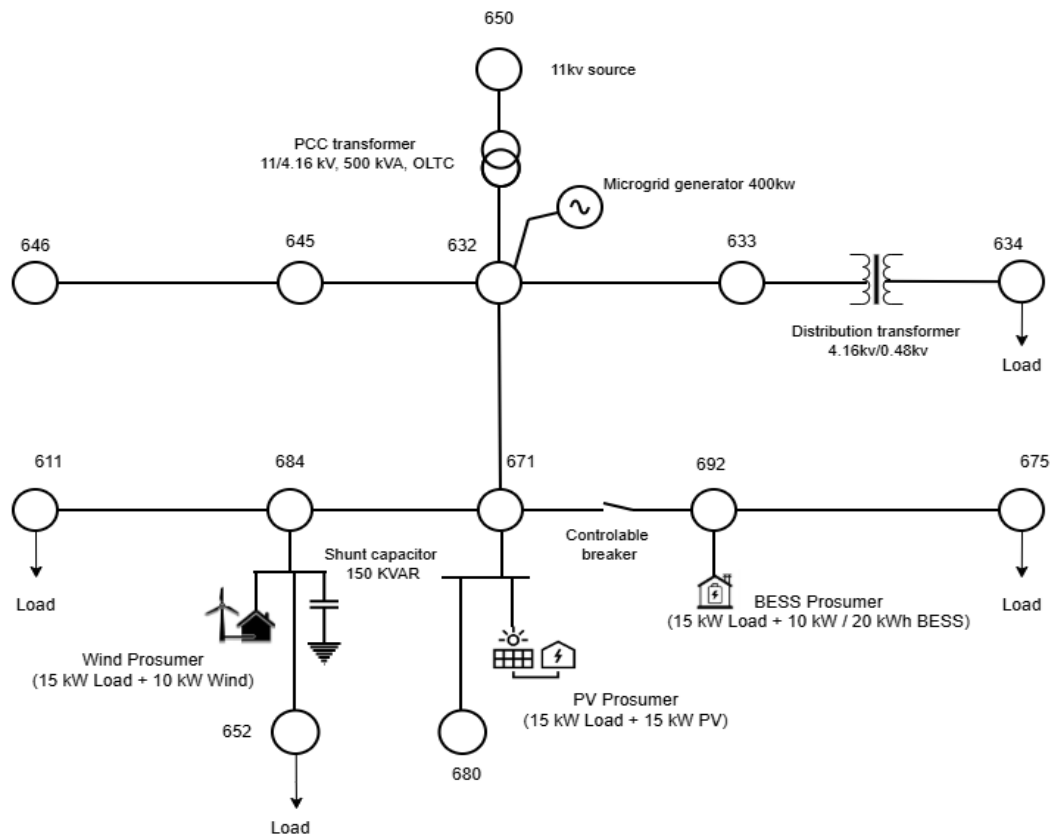


Figure: Single line diagram of modified IEEE 13 bus system